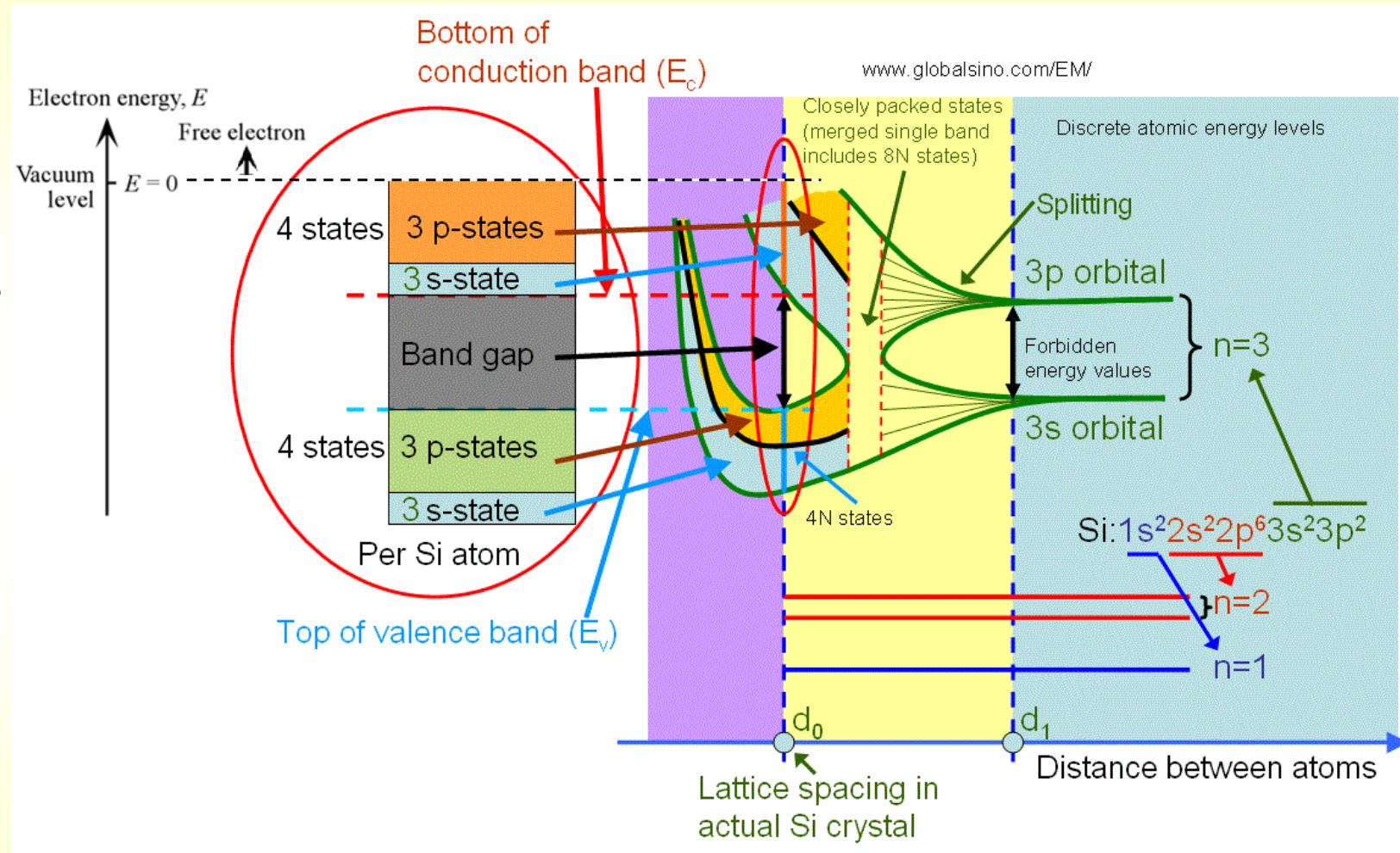


Band theory

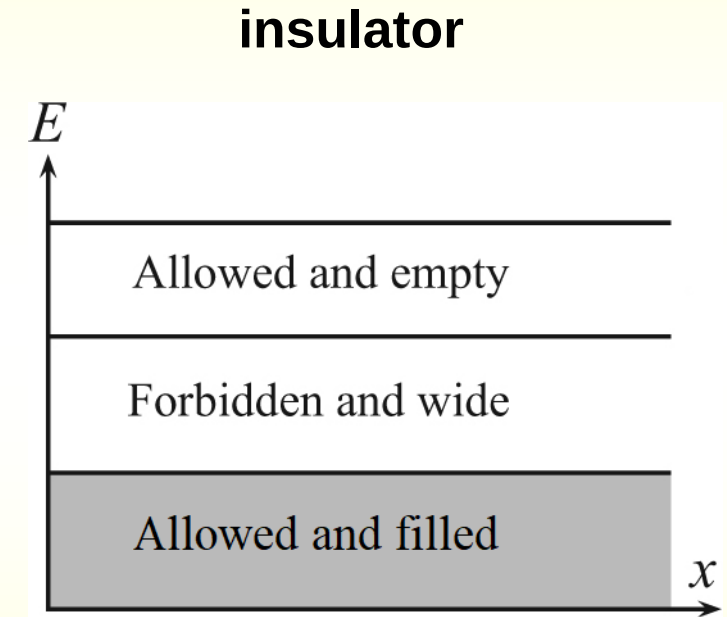
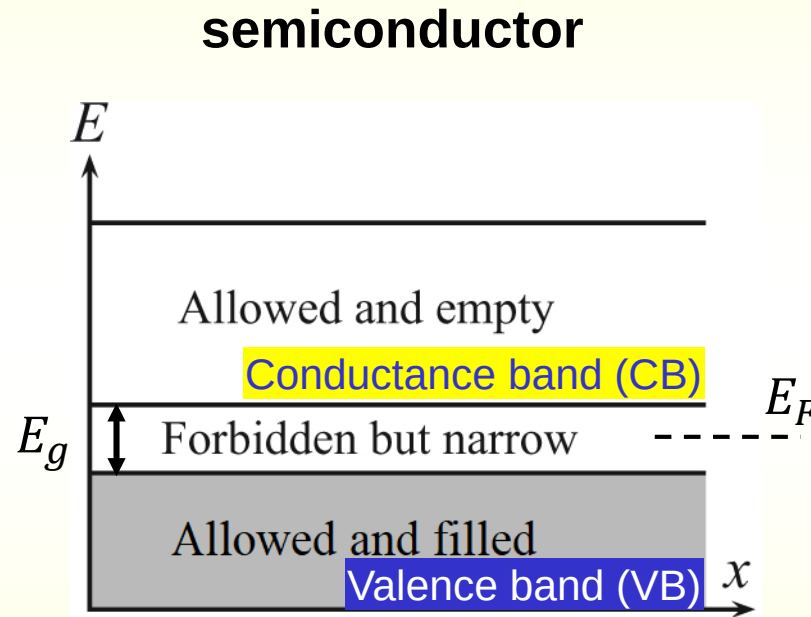
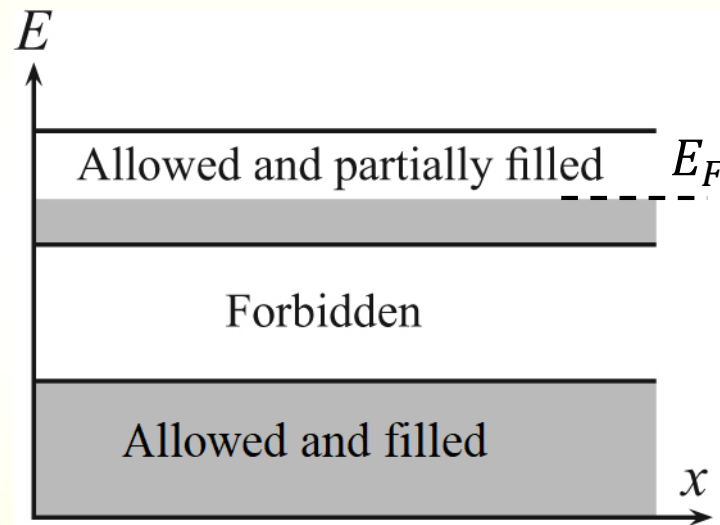
- Energy level splitting vs. atomic distance due to the Pauli's Exclusion Principle





Band theory

- Energy band structure:

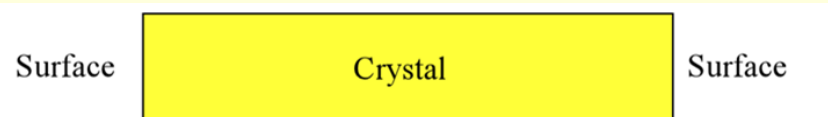
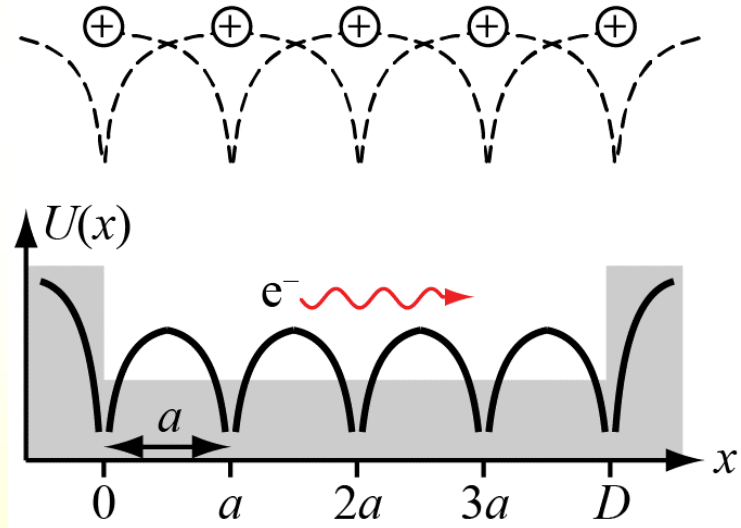


- Fermi energy level:** E_F separates completely filled and completely empty states (at 0 K)
- Band gap energy:** E_g width of forbidden band



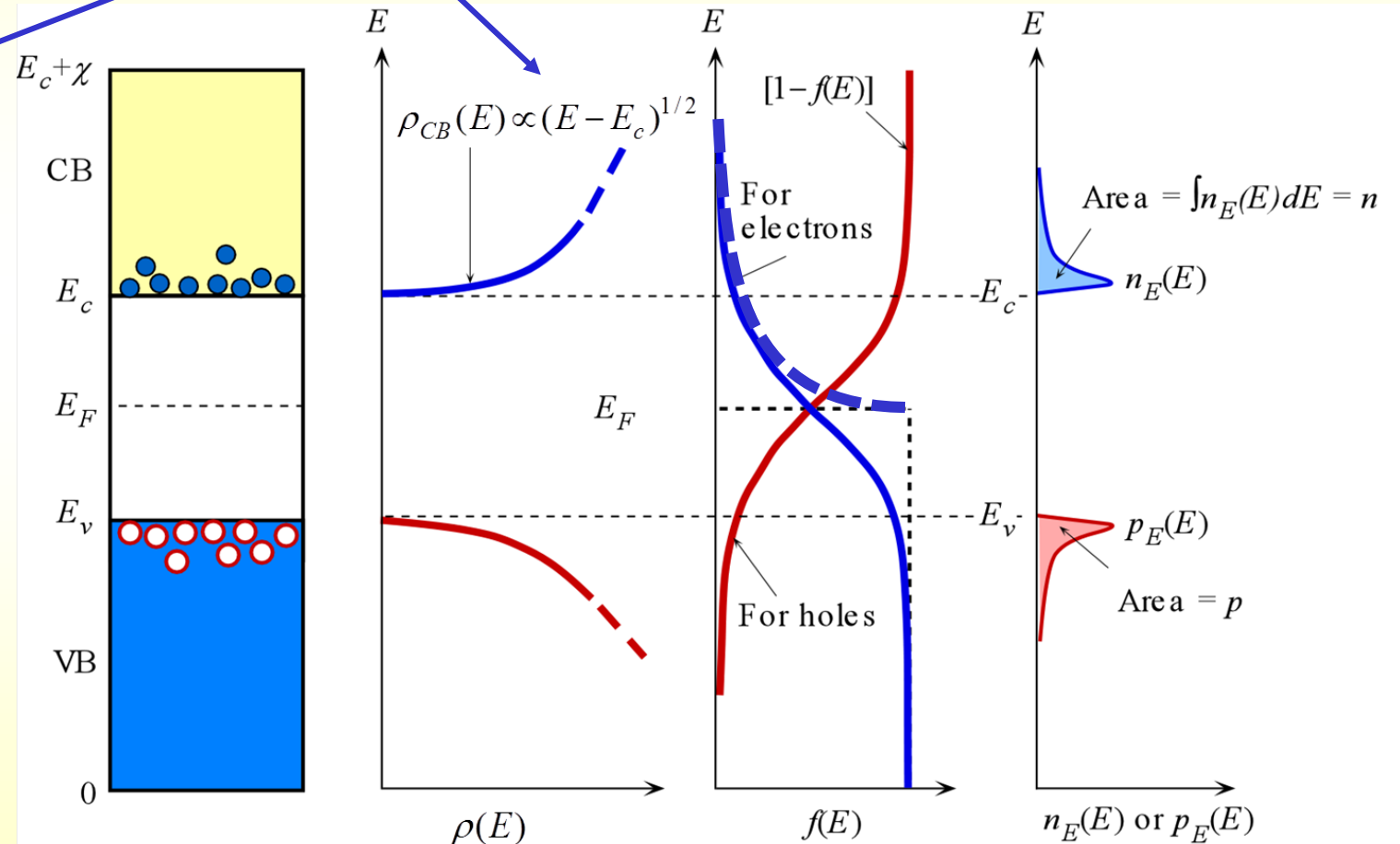
Semiconductors

- Occupation f of available states ρ :
- Free electrons in a box
- Electron wave nature: $p_c = \hbar k$



$$\text{CB: } E = E_c + \frac{p_c^2}{2m_c}$$

$$f(E) = \frac{1}{1 + \exp \frac{E - E_F}{k_B T}} \approx \exp \frac{(-\Delta E)}{k_B T}$$



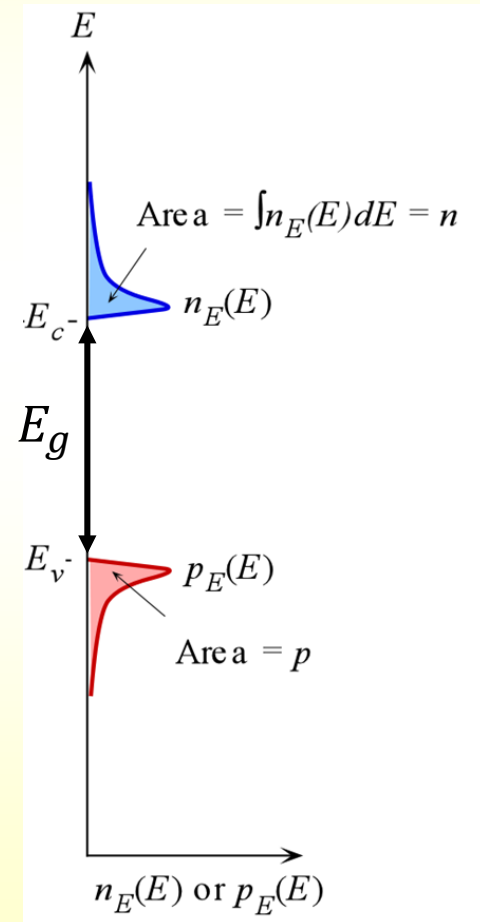


Semiconductors

- Charge carrier concentration in **thermal equilibrium: intrinsic**

$$\left. \begin{aligned} n &= \int_{E_c}^{E_c+\chi} g_{CB}(E) f(E) dE = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \\ p &= \int_0^{E_v} g_{VB}(E) [1 - f(E)] dE = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) \end{aligned} \right\} np = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right)$$

Constant!



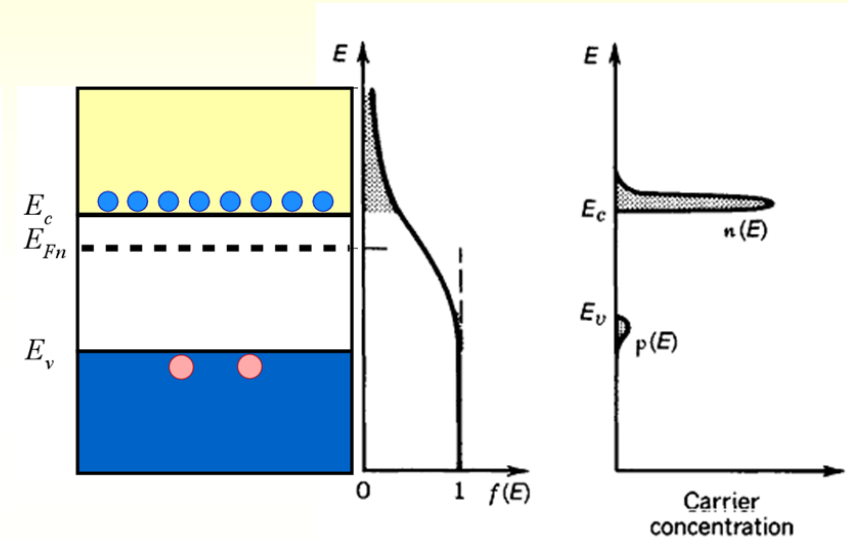
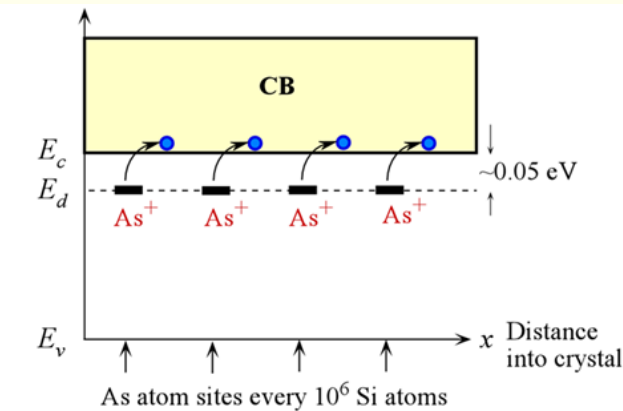
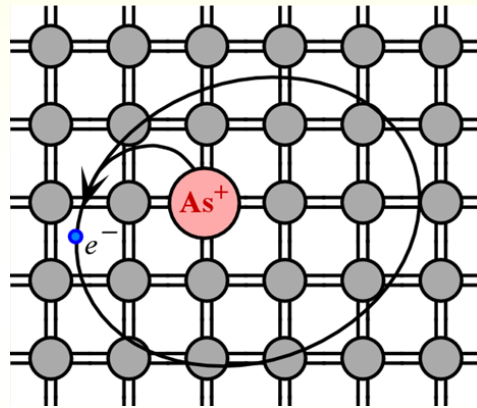
Semiconductors

- Charge carrier concentration in **thermal equilibrium: doped**

- n*-type**

$$n \approx N_d \gg n_i$$

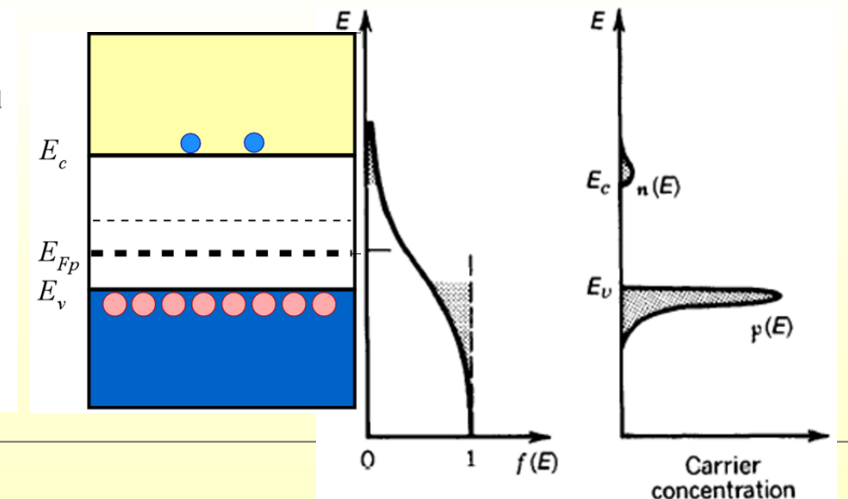
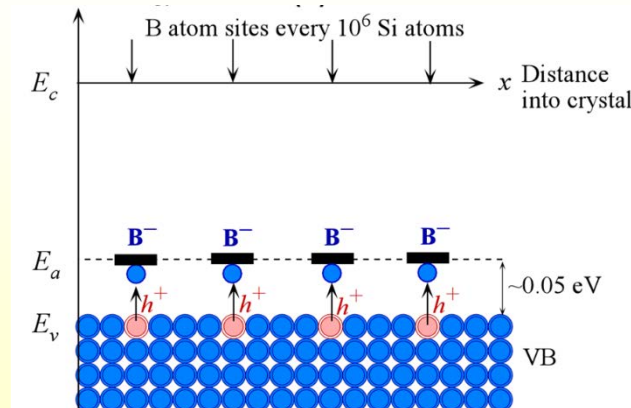
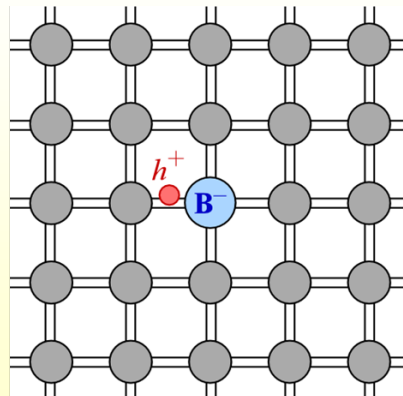
$$p = \frac{n_i^2}{N_d} \ll n$$



- p*-type**

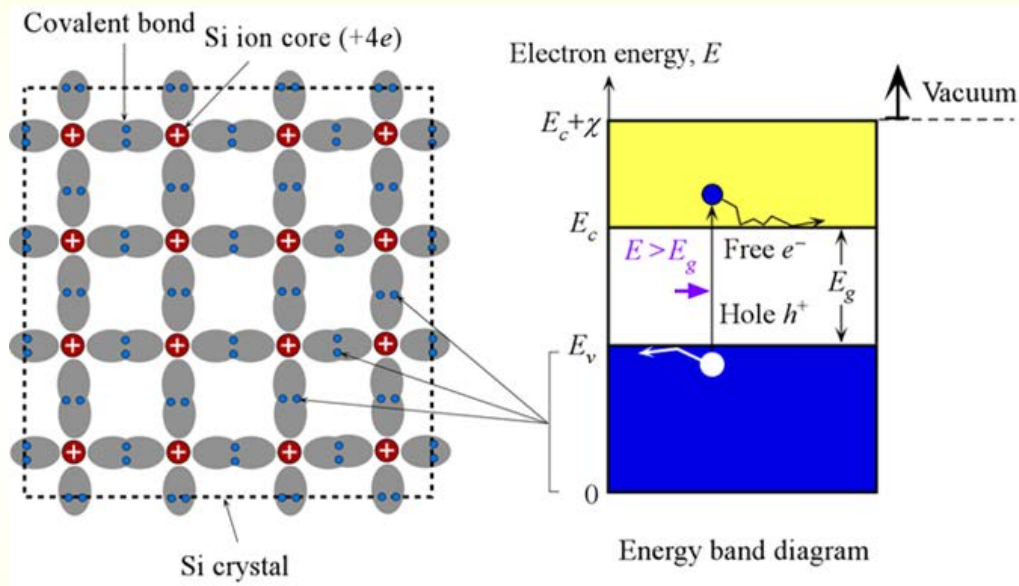
$$p \approx N_a \gg n_i$$

$$n = \frac{n_i^2}{N_a} \ll p$$



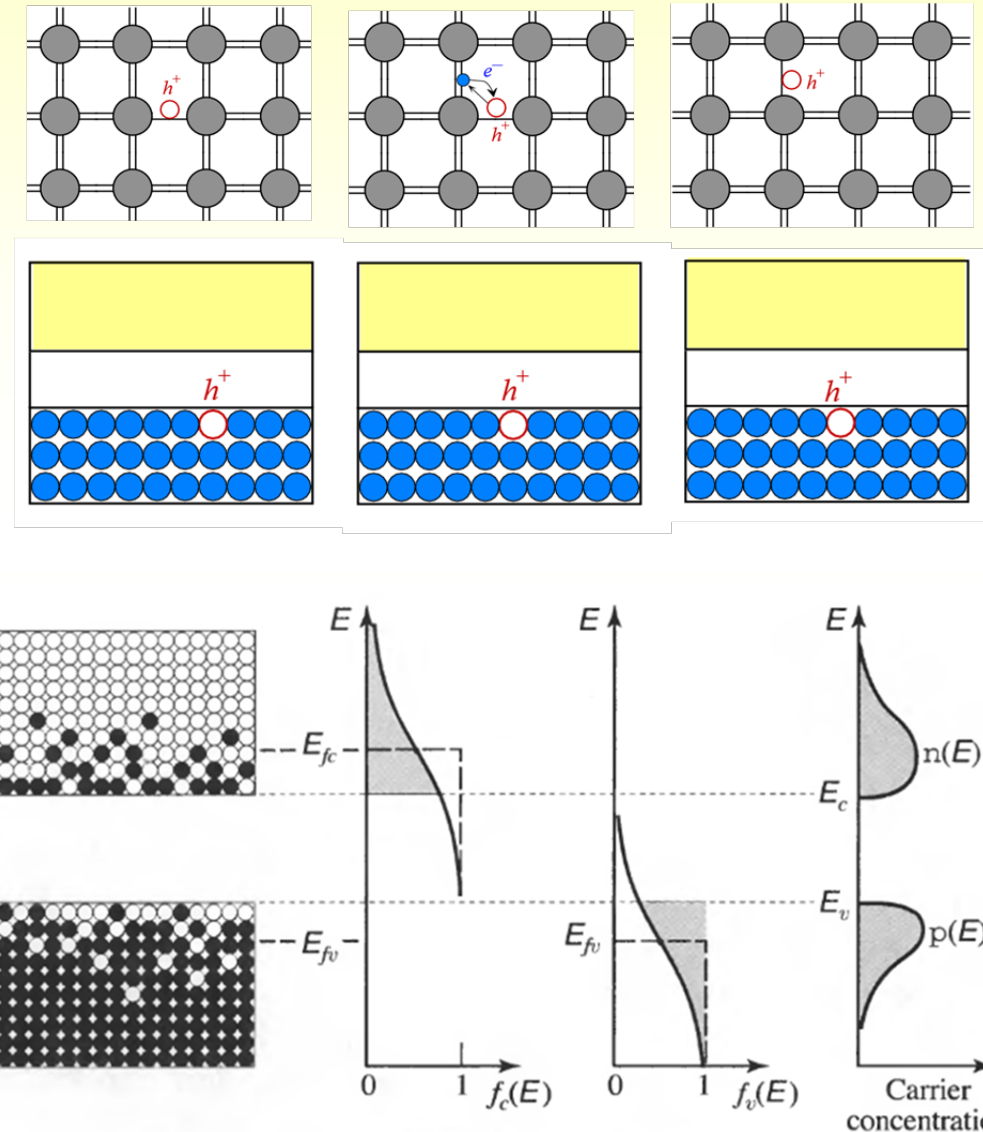
Semiconductors

- Electron-hole pair generation and hole motion



- Quasi-equilibrium carrier concentrations
(due to external electric current or photon flux)

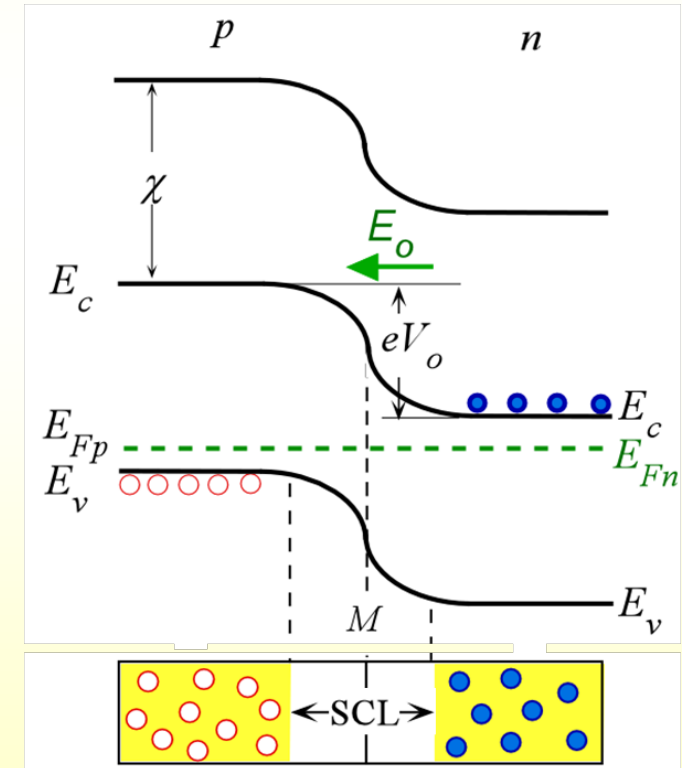
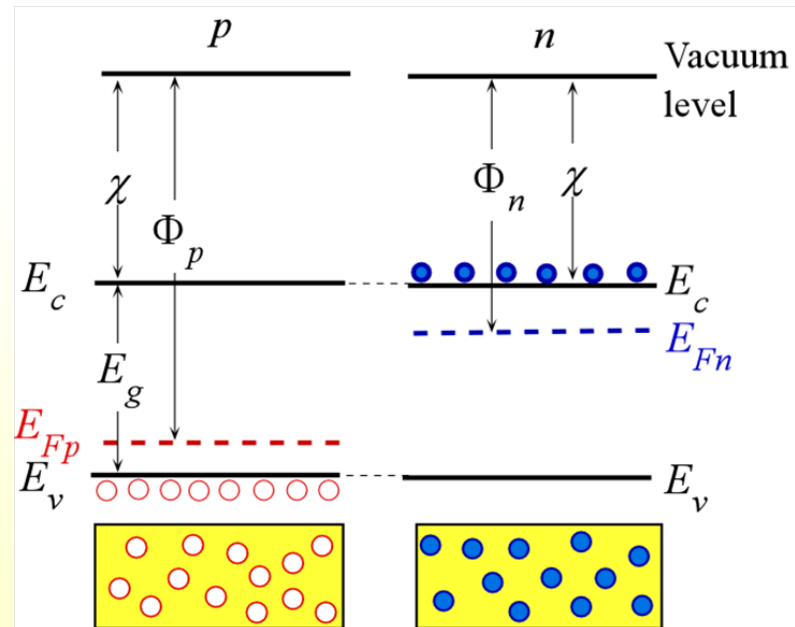
$$np \neq n_i^2$$



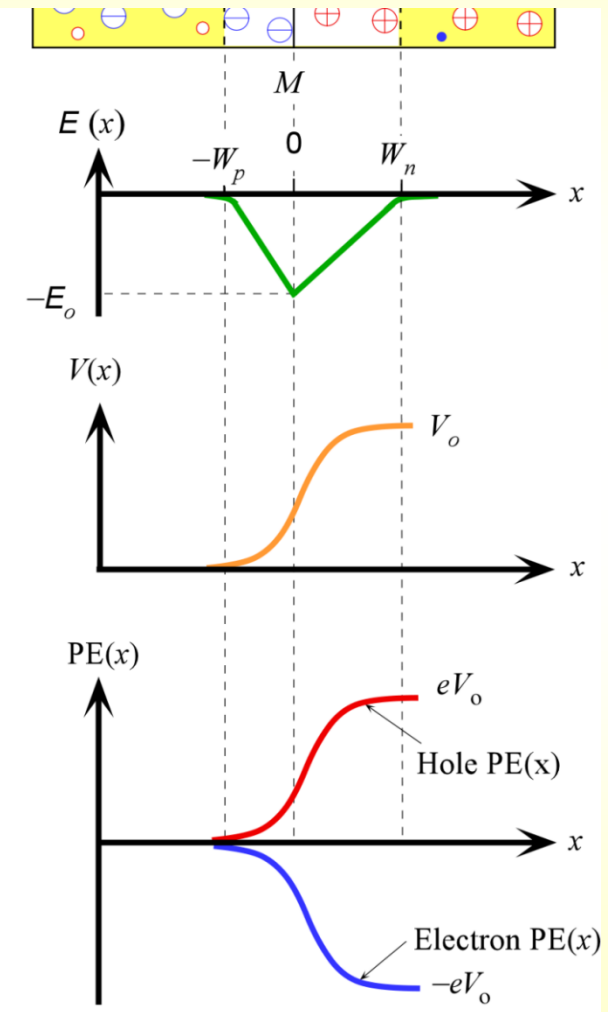
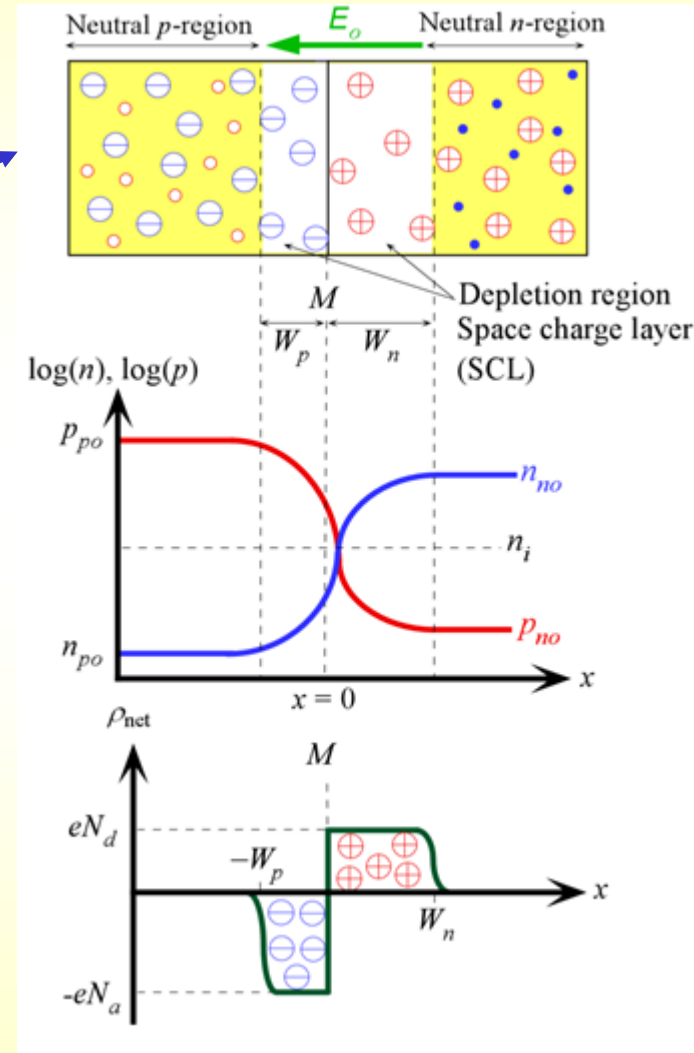
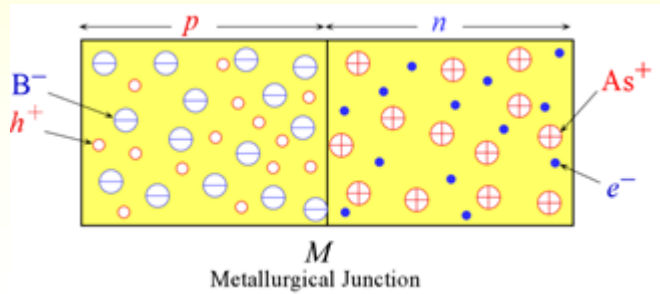


Semiconductor *pn*-junction

- At the boundary of *p*- & *n*-type regions of the same material
- Electrons & holes diffuse toward low concentrations → recombination
→ **depletion layer** with fixed charges of ions → neutrality is not maintained
→ **built-in electric field** → drift & diffusion potentials balance.
- Fermi-levels must align in thermal equilibrium.



Semiconductor *pn*-junction



- Depletion width: $N_a w_p = N_d w_n$
- Space charge density: $\frac{dE(x)}{dx} = \frac{\rho_{net}(x)}{\epsilon_0 \epsilon_r}$
- Electric field: $E_o = \frac{e N_d w_n}{\epsilon_0 \epsilon_r}$
- Potential: $E(x) = -\frac{dV}{dx}$



Semiconductor *pn*-junction

- **Law of the junction** – using Boltzmann statistics

$$\frac{N_2}{N_1} = \frac{f(E_2)}{f(E_1)} \approx \exp\left(-\frac{E_2 - E_1}{k_B T}\right) = \exp\left(-\frac{\Delta E}{k_B T}\right)$$

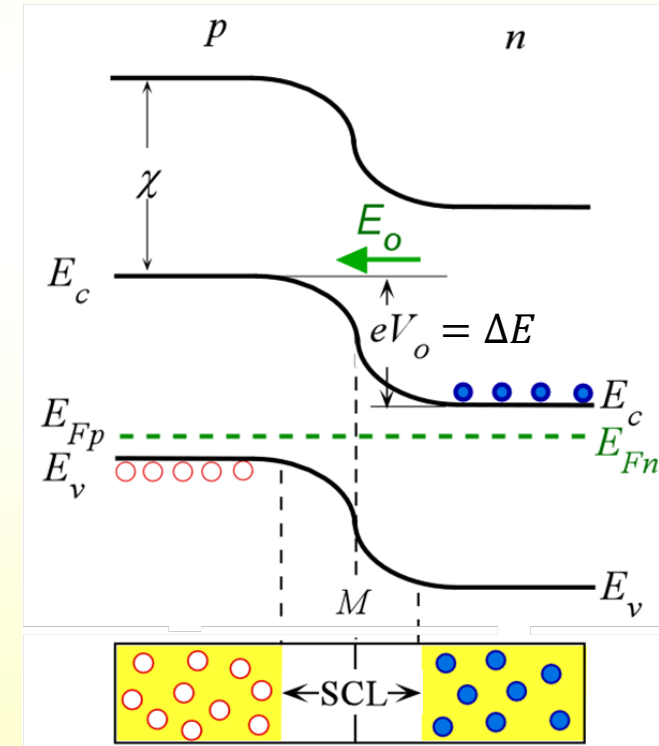
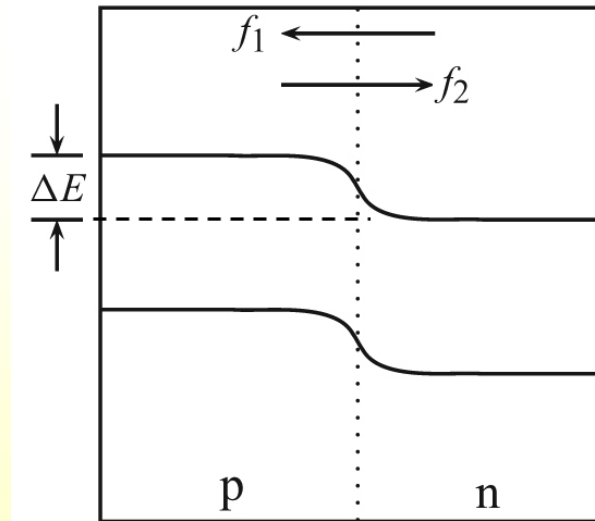
$$\rightarrow \frac{\text{minority}}{\text{majority}} = \frac{n_{p0}}{n_{n0}} = \exp\frac{0 - eV_0}{k_B T} \rightarrow eV_0 = k_B T \ln \frac{n_{n0}}{n_{p0}} = k_B T \ln \frac{N_a N_d}{n_i^2}$$

f_1 : majority electron flow

f_2 : minority electron flow

$$f_1 = C e^{-\frac{\Delta E}{k_B T}} = C e^{-\frac{eV_0}{k_B T}} = f_2$$

$$f_{\text{net}} = f_1 - f_2 = 0$$

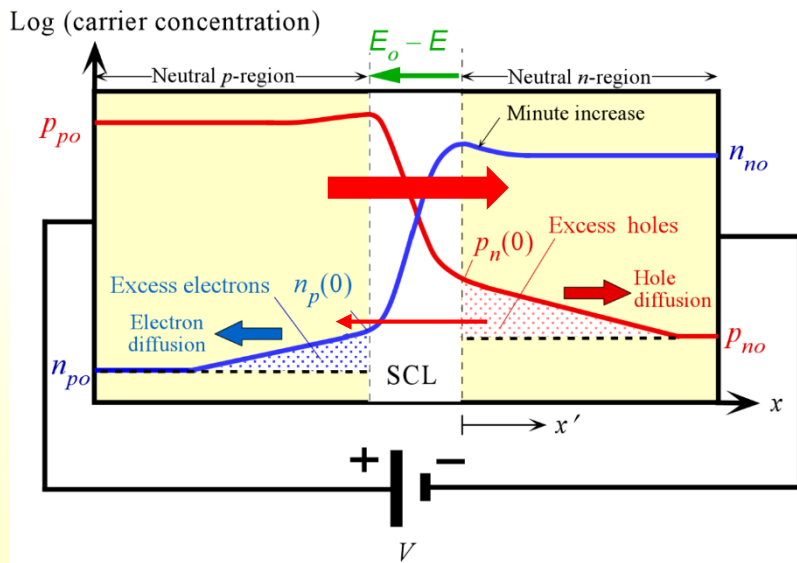


Semiconductor *pn*-junction

- Forward bias:**

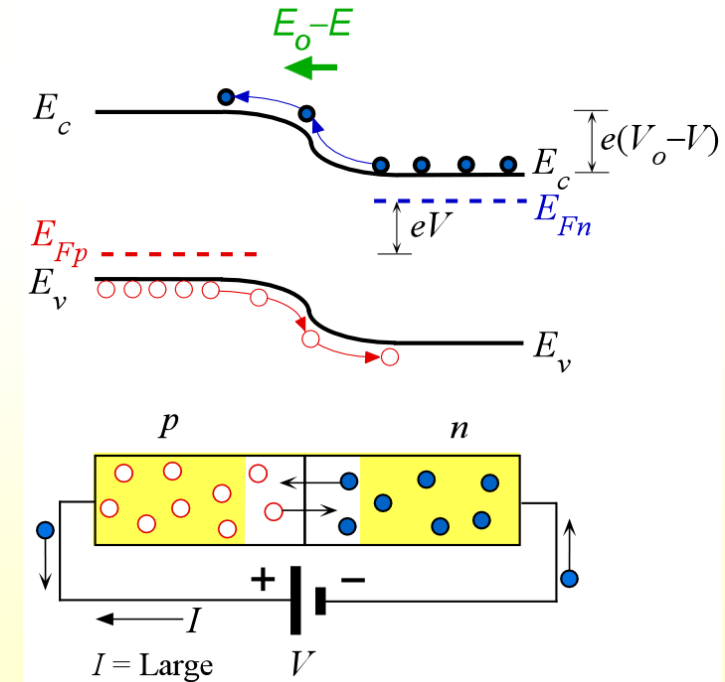
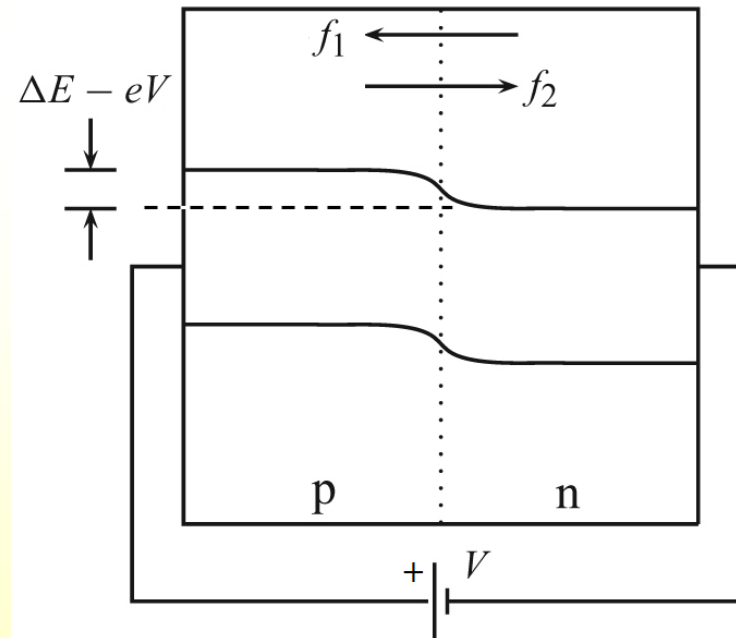
$$\frac{n_p(0)}{n_{n0}} = \exp\left(-e \frac{V_0 - V}{k_B T}\right) = \frac{n_{p0}}{n_{n0}} \exp \frac{eV}{k_B T}$$

$$I_f = I_0 \exp \frac{eV}{k_B T} \quad I_r = -I_0$$



$$f_1 = C e^{-\frac{\Delta E - eV}{k_B T}} \quad f_2 = C e^{-\frac{\Delta E}{k_B T}}$$

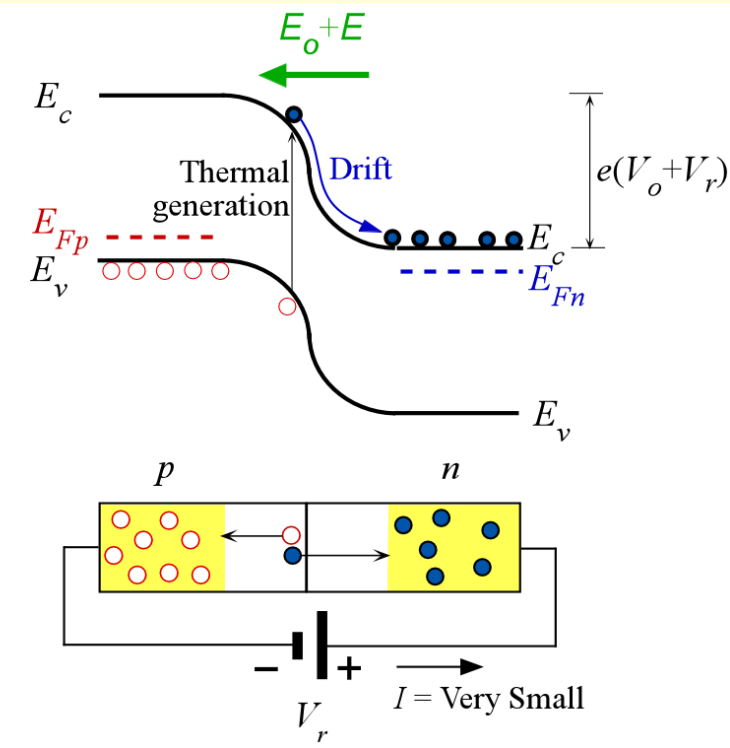
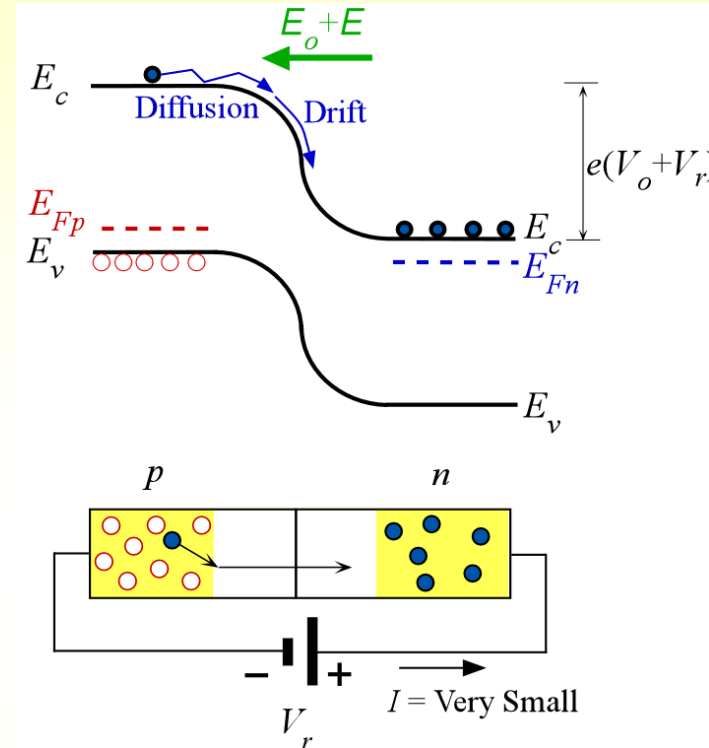
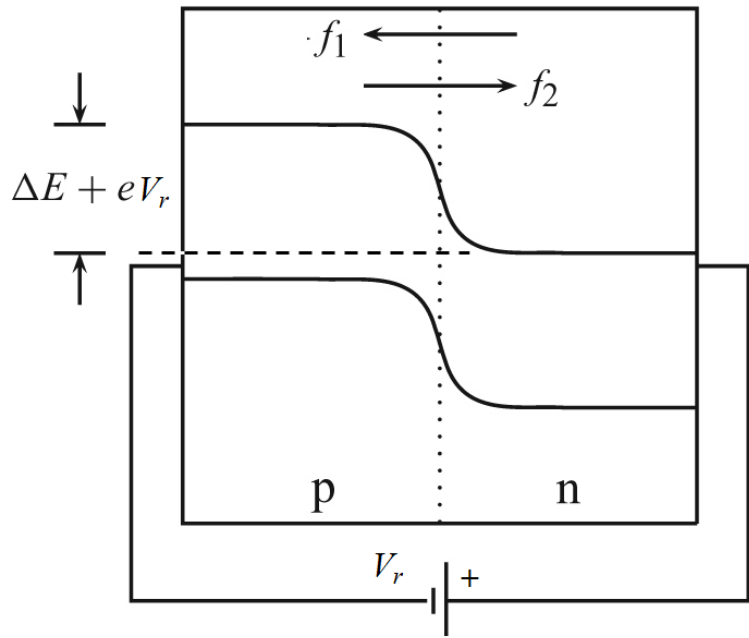
$$f_{\text{net}} = f_1 - f_2 = C e^{-\frac{\Delta E}{k_B T}} \left(e^{+\frac{eV}{k_B T}} - 1 \right)$$





Semiconductor *pn*-junction

- **Reverse bias:**
- Potential barrier increases
- Reverse current



$$f_1 = C e^{-\frac{\Delta E + eV_r}{k_B T}} \quad f_2 = C e^{-\frac{\Delta E}{k_B T}} \quad k_B T \approx 0.025 \text{ eV}$$

$$f_{\text{net}} = f_1 - f_2 = C e^{-\frac{\Delta E}{k_B T}} \left(e^{-\frac{eV_r}{k_B T}} - 1 \right) \approx -C e^{-\frac{\Delta E}{k_B T}}$$

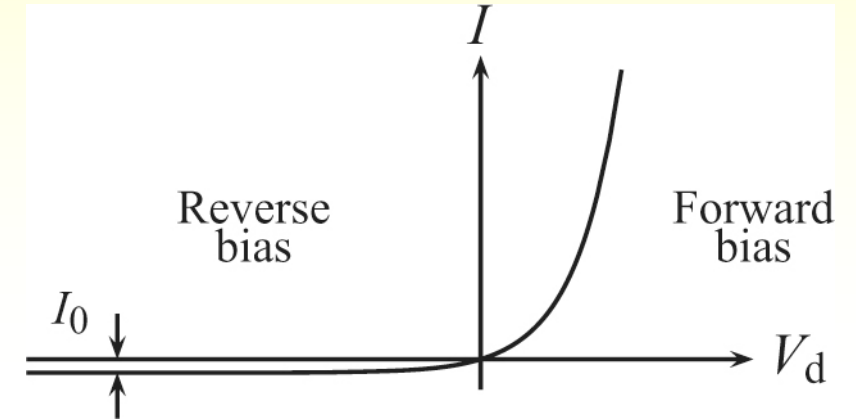
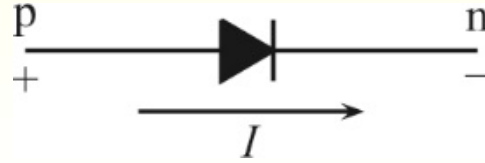


Semiconductor *pn*-junction

- Total diode characteristics

$$I = I_0 \left(\exp \frac{eV_d}{k_B T} - 1 \right)$$

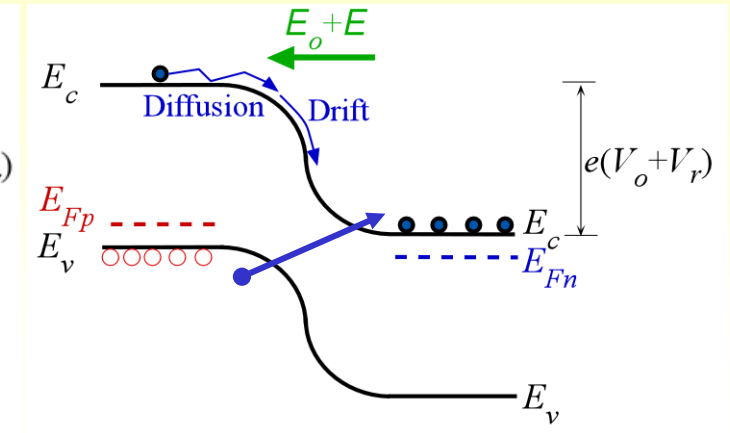
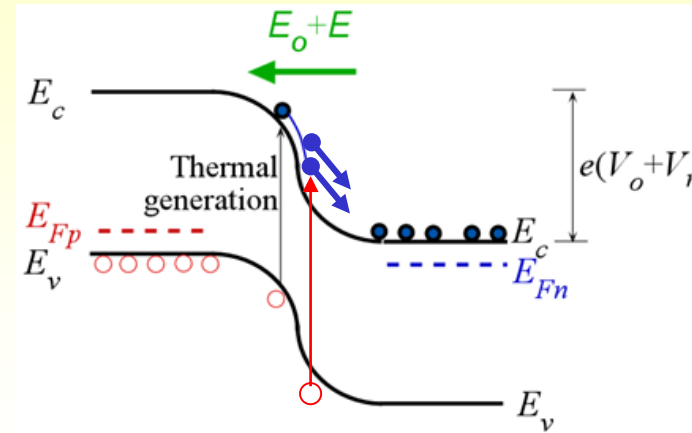
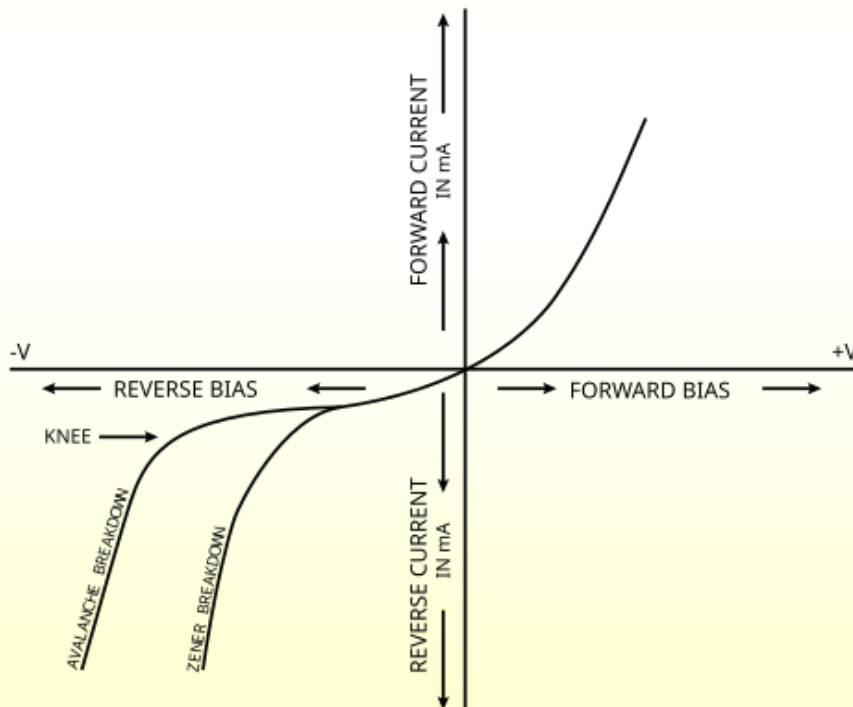
$$I_0 \equiv C e^{-\frac{\Delta E}{k_B T}}$$





Semiconductor *pn*-junction

- **Breakdown:**
- Avalanche (impact ionization)
- Zener (direct ionization + tunnelling)



Avalanche Breakdown

- > It occurs in junctions which are lightly and have wide depletion widths.
- > It occurs at higher reverse voltages when thermally generated electrons get enough kinetic energy to produce more electrons by collision.
- > At reverse voltage above 6V breakdown is due to avalanche effect.
- > Charge carriers obtain energy from the applied potential.

VS

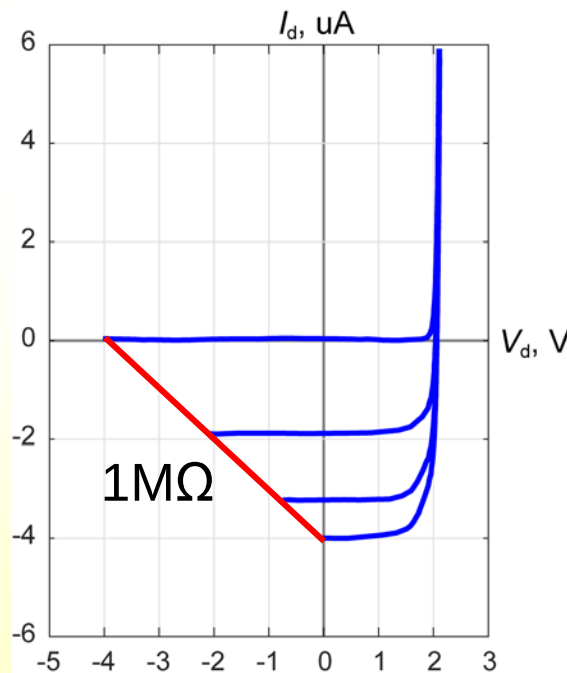
Zener Breakdown

- > It occurs in junctions which are heavily doped and have narrow depletion widths
- > It occurs due to rupture of covalent bonds by strong electric fields set up in depletion region by the reverse voltage.
- > At reverse voltage below 6V breakdown is due to zener effect.
- > Zener current is independent of applied voltage.

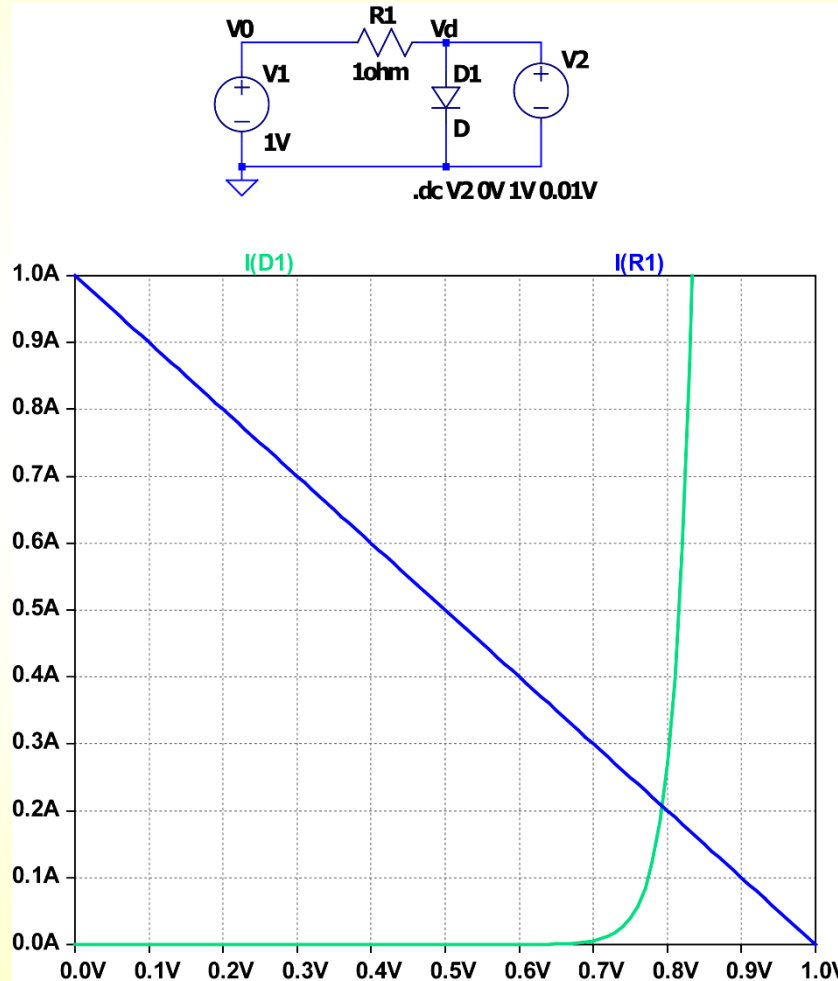


Diode characteristics

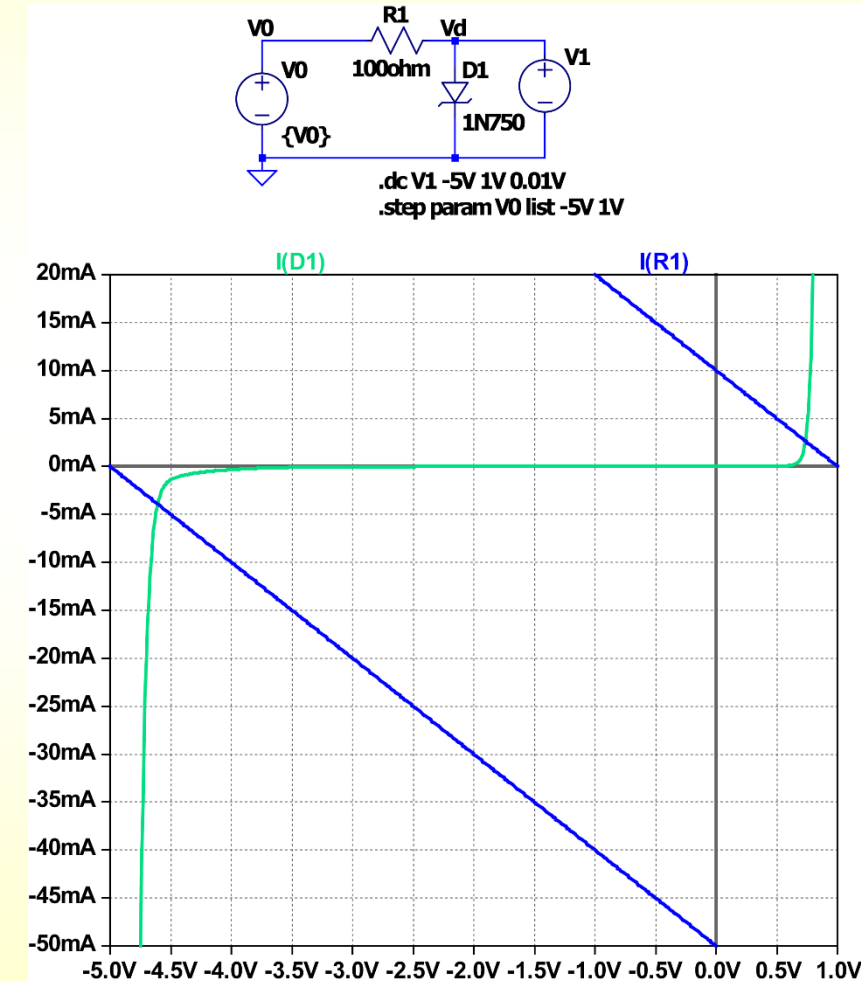
- Load line and operating point
- Photodiode (reverse biased)



Forward biased diode



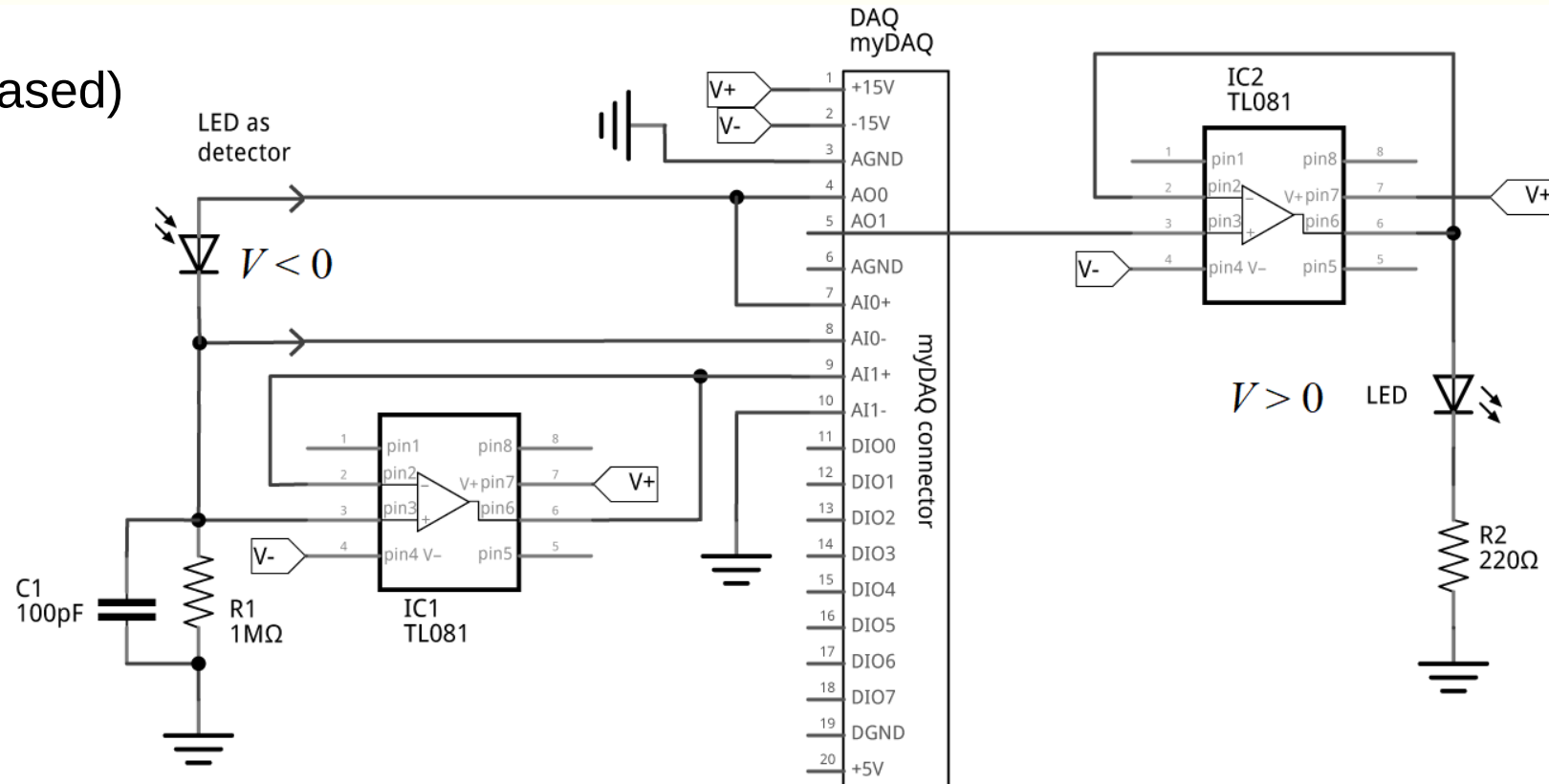
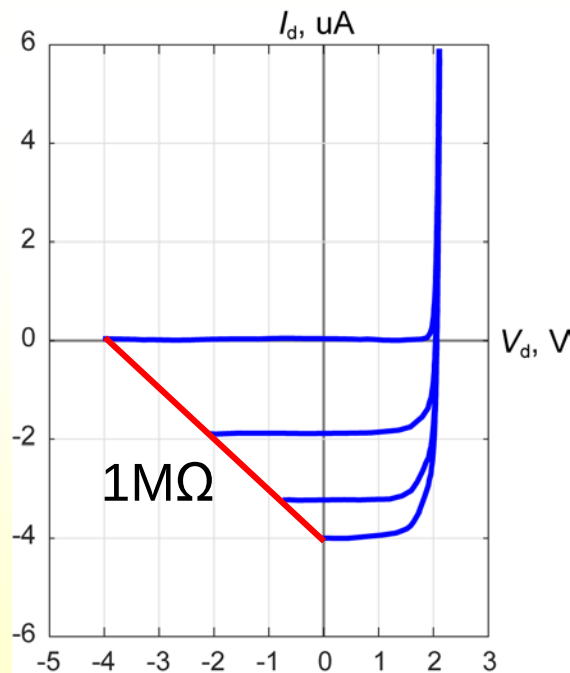
Zener diode





Diode characteristics

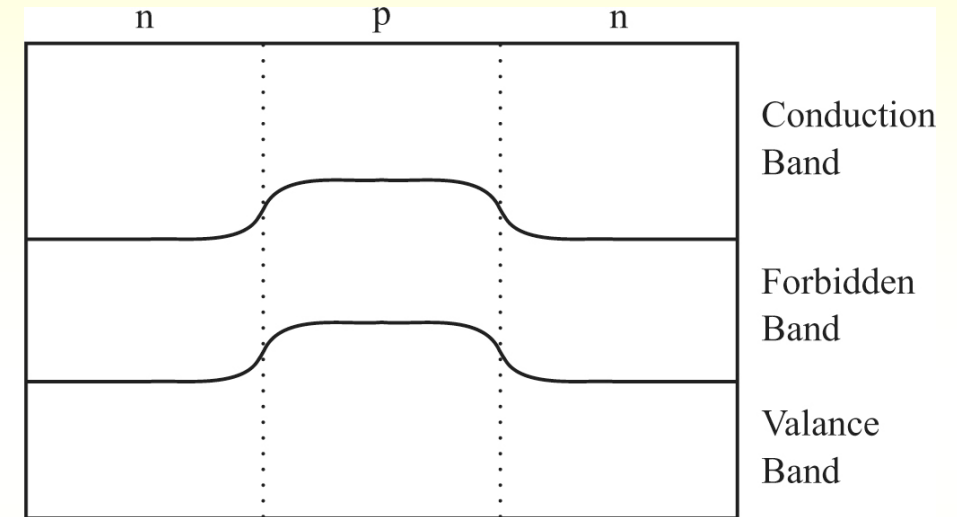
- Load line and operating point
- Photodiode versus LED
(reverse biased) (forward biased)



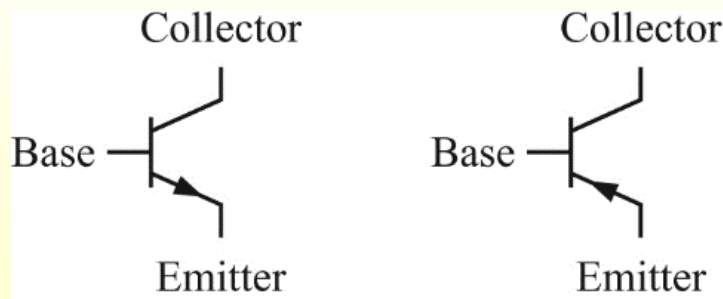


Bipolar junction transistor

- Unbiased **NPN**



- NPN** and **PNP** symbols



emitter:
heavily
doped

base:
lightly
doped

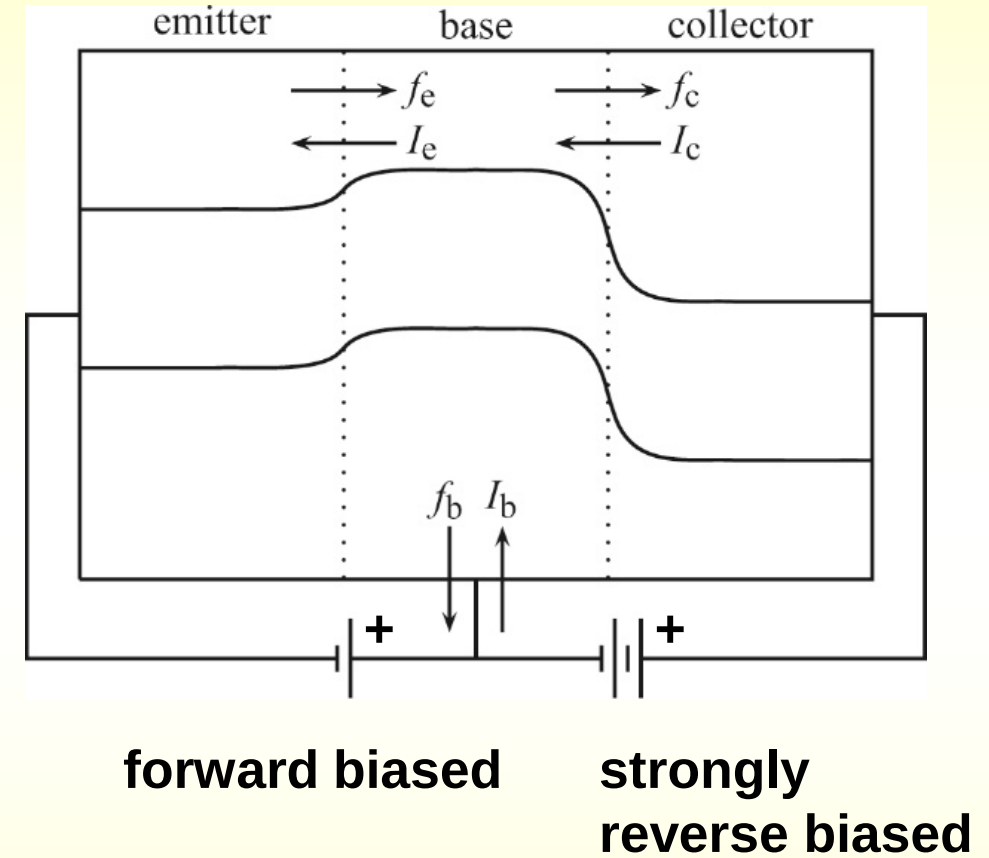
collector:
lightly
doped



Bipolar junction transistor

- **NPN** transistor biased for **linear active operation**

- 1) Forward biased **E-B** junction (**E** heavy doped): net flow of electrons to **B** (f_e)
- 2) Reverse biased **C-B** junction: very little flow of electrons into **B**
- 3) Electrons from **E**:
 - a) Recombine with a hole in **B**. An electron must flow out of **B** to maintain charge neutrality. Otherwise **B** quickly charges up negatively and stops f_e .
 - b) Pass through **B** into **C**, leading to electron flow f_c out of **C**. Thin, lightly doped **B** promotes this by reducing hole concentration. Light **C** doping increases depletion width.





Transistor characteristics

- $I_b(V_{be})$ and $I_c(V_{ce})$
- Load line and operating point
- Small signal AC:
 $\rightarrow V_1 + v_1(t)$
- Linear model

